



Storyboarding and Low-Fidelity Prototypes

Name: Rishabh Ray

Student ID: 34525416

Table of Contents

1. Introduction
2. Selected User Stories
3. Storyboards
4. Kanban Boards
5. Low-Fidelity Prototype Screens
6. Design Report
7. Conclusion
8. Appendix – Persona





Introduction

This report details the design process for a mobile application aimed at solving a critical challenge for childcare manager Arjun Mehta : the need to efficiently source safe, hygienic toys in bulk while building trust with donors.

Guided by two prioritised user stories, the design progresses from contextual storyboards that explore Arjun's challenges to measurable acceptance criteria and three low-fidelity prototype screens. These prototypes implement the core requirements and are annotated to demonstrate the application of key usability and accessibility principles.

The report concludes by identifying the limitations of this design stage and proposing specific goals for future high-fidelity development.



2. Selected User Stories



Story 1 – Must Have

As Arjun, I want to collect safe and hygienic toys in bulk so that I can provide enough play and learning materials for all children in my childcare centre.

Story 2 – Should Have

As Arjun, I want to build trust with parents and donors so that I can consistently receive reliable toy contributions for my childcare centre.

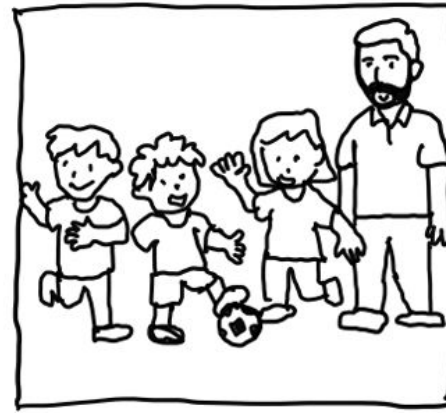
Justification: No changes made. Both stories remain relevant, validated by survey data on hygiene concerns



3. Storyboards



Storyboard A – Bulk Safe Donations (Story 1)



Storyboard B – Trust with Donors (Story 2)



Arjun at childcare centre



The dashboard shows donor history and trust badges



Voice-to text chat allows me to communicate without typing



Chat builds direct trust and reassurance



Transparent feedback builds the donor's reputation





4. Kanban Boards

A detailed close-up of a printed circuit board (PCB) during a soldering process. A soldering iron is shown in the lower right, its tip touching a component. The board is populated with various electronic components, including a large black electrolytic capacitor labeled '132400' and several resistors labeled '475'. Other labels like 'R4', 'R5', 'R3', and 'C4' are visible. The background is a dark, solid color, and the title '4. Kanban Boards' is overlaid in white text.



Story 1 – Bulk Safe Donations

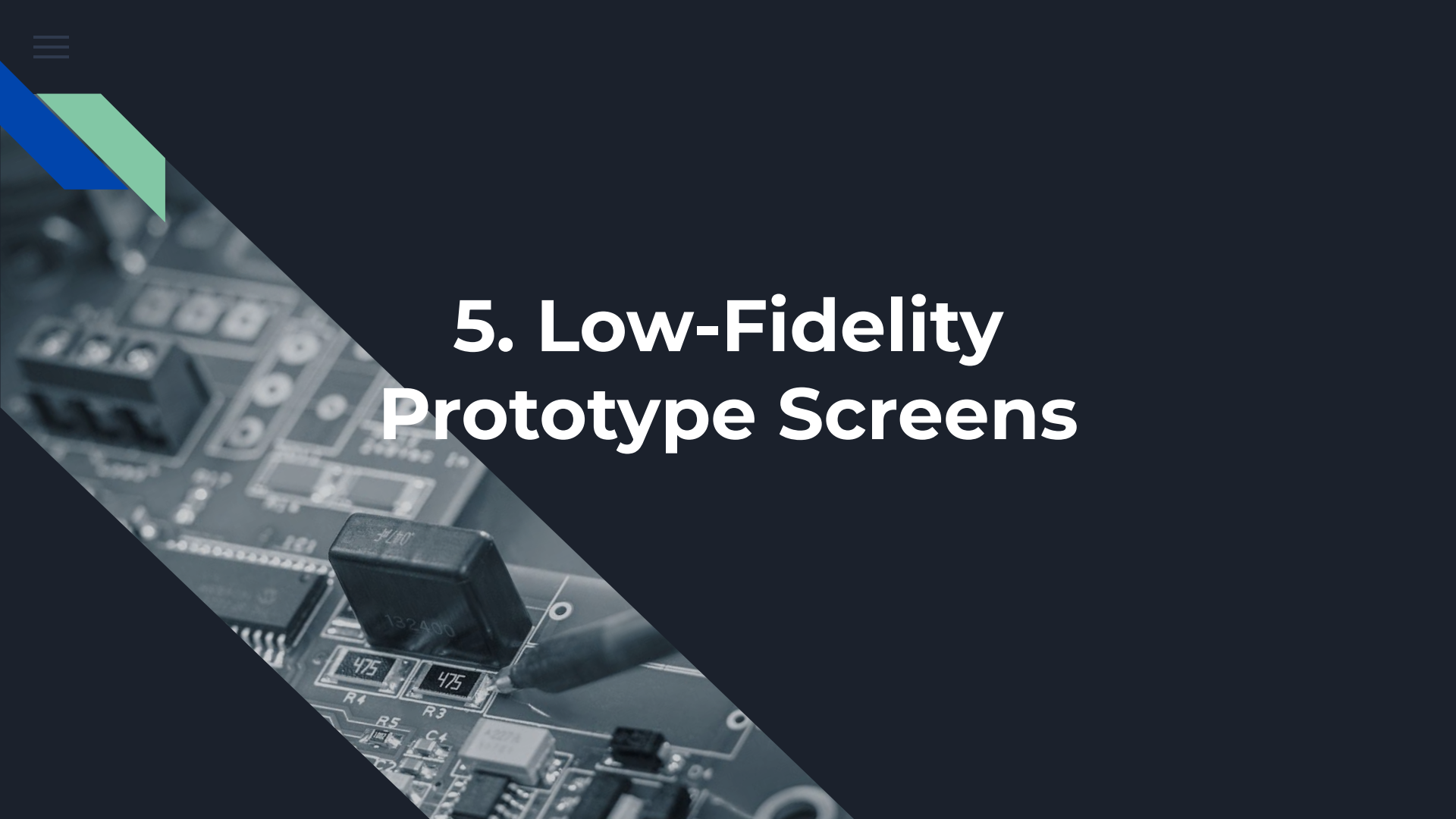
User Story	To Do	Doing	Done
<i>As Arjun, I want to collect safe and hygienic toys in bulk so that I can provide enough play and learning materials for all children in my childcare centre.</i>	Allow multiple toys to be uploaded in a single donation batch. Require condition, hygiene, and age suitability fields for each toy.	Display a “Verified Clean/Safe” tag before acceptance.	Group bulk donations by donor for efficient intake. Provide confirmation message once donation submitted.

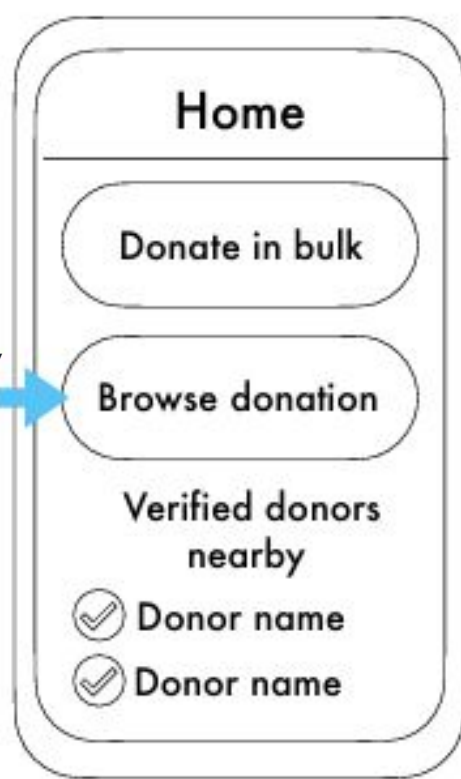
Story 2 – Trust with Donors

User Story	To Do	Doing	Done
<i>As Arjun, I want to build trust with parents and donors so that I can consistently receive reliable toy contributions for my childcare centre.</i>	Donor profiles must show verified history of past donations. Trust badges auto-display after ≥ 3 successful donations.	Recipients can leave feedback on each donation.	Donation history accessible from dashboard. Voice to Text chat accessibility feature

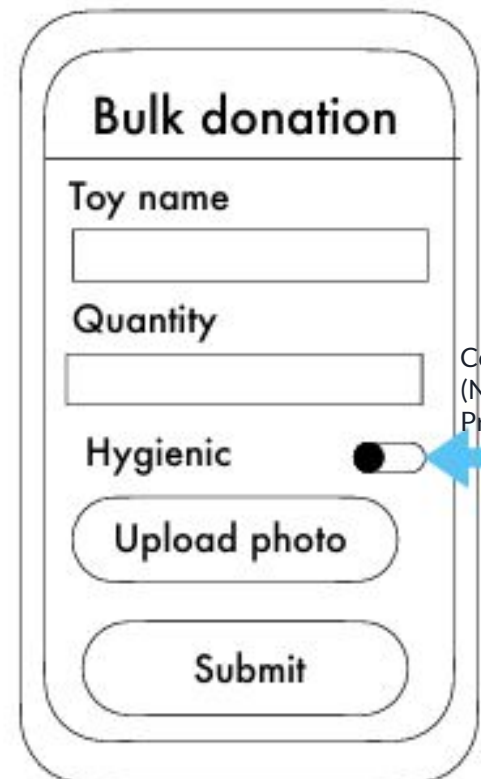


5. Low-Fidelity Prototype Screens

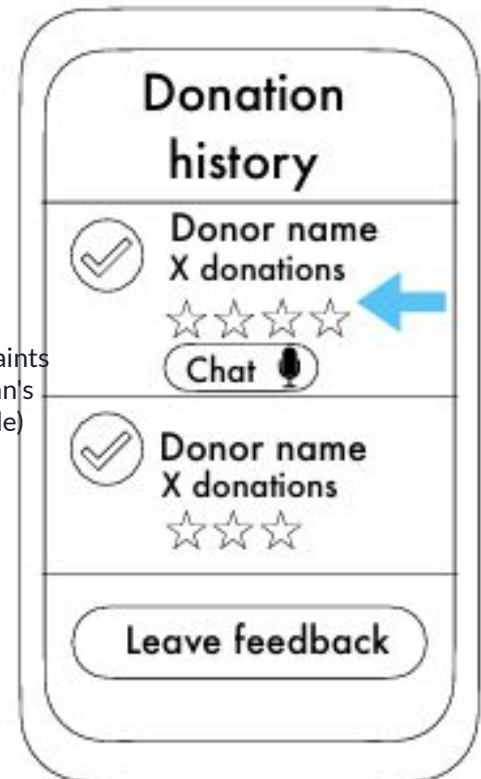
A detailed close-up of a printed circuit board (PCB) during assembly. A soldering iron is shown in the process of soldering a component. The board is populated with various electronic components, including a large black electrolytic capacitor labeled '132400', several resistors with values like '475', and integrated circuits. Labels such as 'R4', 'R5', 'R3', and 'C4' are visible on the board. The background is a dark, solid color.



Operable



Perceivable



Understandable

Discoverability
(Norman's
Principle)



Constraints
(Norman's
Principle)



Feedback
(Norman's
Principle)





6. Design Report



Screen 1 – Home Dashboard

User Story: This screen acts as the entry point for User Story 1 (collecting safe toys in bulk) and User Story 2 (building trust with donors)

Norman Principle (Discoverability): The home screen applies the principle of Discoverability by presenting the main user actions, "Donate in bulk" and "Browse donation," as visible buttons. As shown by the annotation in the sketch, these clear signifiers make the app's options immediately identifiable and reduce cognitive load.

Accessibility Principle (Operable): The design ensures the interface is Operable. The buttons and tap targets are drawn to be large, supporting one-handed use. This is critical for the persona, Arjun, who may be navigating the app while supervising children, thus addressing potential situational impairments and preventing exclusion.



Screen 2 – Bulk Donation Form

User Story: This screen is designed to support User Story 1, which focuses on collecting safe and hygienic toys in bulk.

Norman Principle (Constraints): The form applies the principle of Constraints through the "Hygienic" field. As indicated by the annotation, this toggle switch constrains the user's input, preventing unsafe or incomplete submissions and enforcing the safe toy standards required for a childcare setting.

Accessibility Principle (Perceivable): The design ensures information is Perceivable for users. For instance, the hygiene status is conveyed using both a text label ("Hygienic") and an icon (the toggle switch). This approach ensures the status is perceivable by users with vision strain and does not rely on color alone.



Screen 3 – Donation History & Trust Dashboard

User Story: This screen supports User Story 2 by providing features to build trust between recipients and donors.

Norman Principle (Feedback): The design provides clear Feedback through elements like trust badges (the checkmark icon) and feedback stars. As the annotation shows, these features offer immediate and informative feedback on past donations. This helps build confidence in donors and creates transparency for recipients like Arjun.

Accessibility Principle (Understandable): The interface is designed to be Understandable. It uses simple language and symbols, such as "Chat" and "Leave feedback," and maintains a consistent layout with a clear list of donors. This reduces complexity and ensures the dashboard is easy to use, even in a distracting childcare environment.



7. Conclusion

These deliverables demonstrate how Arjun's needs are translated into usable prototypes. Storyboards contextualised his challenges, while Kanban boards made acceptance criteria measurable. The prototype sketches efficiently implemented these criteria using key design and accessibility principles.

However, low-fidelity sketches cannot validate key interactive elements. Therefore, the first goal for high-fidelity prototyping is to

(1) conduct live user tests to measure completion rates for the hygiene-field constraints.

Similarly, static sketches cannot gauge emotional trust, so a second goal is to

(2) evaluate whether interactive trust badges and donor histories tangibly increase user confidence and reliability.

8. Appendix – Persona (from Submission 1)

ARJUN MEHTA



“Every child deserves safe, joyful play — no matter where the toy comes from.”

AGE 32

OCCUPATION Childcare Centre Founder

HOUSEHOLD Married with children

LOCATION Melbourne, VIC

ARCHETYPE Community Builder

Practical

Sustainability-Focused

Child-Safety Advocate

Community-Driven

Biography

Arjun is the founder of a small childcare initiative in Melbourne. As a father himself, he knows the importance of safe, engaging toys for children's growth. Rising childcare costs and limited resources pushed him to find creative solutions — leading him to bulk toy donations.

For Arjun, pre-loved toys aren't second-hand, they're opportunities: affordable, sustainable, and meaningful when given new life in a child's hands. He actively connects with local parents, schools, and online donation groups, often juggling the dual roles of organiser and caregiver. Trust is central to his work — every toy must be safe, clean, and worthy of a child's playtime.

Behavioural Trait

SAFETY-FIRST: Checks every toy before use, prioritising hygiene and safety.

RESOURCEFUL COLLECTOR: Proactively seeks bulk donations to stretch limited funding.

CHILD-CENTRIC THINKER: Selects toys based on educational and developmental value, not just availability.

VARIETY SEEKER: Values having a wide range of toys to match children's learning needs.

Motivations & Goals

- Secure bulk toy donations to reduce costs and ensure continuous supply.
- Ensure all donated toys meet hygiene and safety standards.
- Build trust with parents and donors.
- Provide a diverse range of play and learning opportunities.
- Promote sustainability by giving pre-loved toys a second life.

Digital Habits

Community-Platform User:



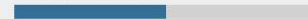
Visual Communicator



Tech Explorer



Privacy Awareness



Frustrations & Pain Points

- Difficulty verifying cleanliness and safety of donated toys.
- Limited funding to purchase toys new.
- Inconsistent supply of donations in bulk.
- Time constraints from balancing childcare management and donation sourcing.
- Rising demand from families can sometimes exceed available resources.

Persona: Arjun Mehta (32 years old, Childcare Centre Manager)